Forecasting India's Eighth Pay Commission and Its Household Consumption Impacts: A Hybrid Econometric–Machine Learning Approach

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Abstract— **This paper introduces a comprehensive, hybrid econometric–machine learning framework designed to forecast the primary parameters of India's forthcoming Eighth Pay Commission and to quantify their consequent effects on household consumption expenditure across rural and urban sectors. Drawing on data from the 3rd through 7th Pay Commissions, we first apply ARIMA and SARIMAX techniques to project macroeconomic drivers—GDP per capita, net national income, and inflation. Next, support vector regression estimates a disposable income index, while polynomial and ordinary least squares regressions model fiscal impact and real pay increases. A multi-output ridge regression then predicts minimum and maximum pay hikes, which, together with projected macro drivers, feed into a final ridge regression stage that disaggregates per-capita consumption into food and non-food categories for rural and urban populations. Our analysis forecasts pay revisions of approximately +171.3% (minimum pay) and +202.3% (maximum pay), generating an aggregate fiscal burden near 2.13 trillion. We find that non-food expenditure surges by over 250% in urban areas and by about 11% in rural regions, with elasticity analyses highlighting strong stimulus in urban clothing, footwear, and durable goods, and in rural beverages. We discuss policy implications—such as phased pay adjustments, targeted subsidies, and monetary safeguards—to balance welfare gains with fiscal stability. Limitations include the small number of historical observations and data gaps, suggesting future work on integrating micro-survey data, deep-learning nowcasting, and scenario-based stress tests. This replicable framework equips policymakers, committees, and private stakeholders with a data-driven decision-support tool for designing equitable, sustainable pay revisions.**

I. Introduction

Since 1946–47, the Government of India has appointed successive Pay Commissions to review and revise public-sector remuneration, balancing cost-of-living adjustments, pay-scale compression, and fiscal sustainability [1]. Each Commission's recommendations—from the Third through the Seventh—have driven average real pay increases of 15–25% and altered compression ratios, reshaping both salary structures and long-term pension liabilities [2]. These revisions raise disposable incomes, influence per-capita national income, and can exert upward pressure on inflation and fiscal deficits, underscoring the need for robust forecasting of future liabilities.

Applying forecasting and machine-learning techniques to this domain presents several challenges. Data scarcity and granularity limit our time-series length—only five historical Commissions align with quinquennial NSSO/HCES surveys (1993–94 to 2009–10) [4]. Structural breaks, such as economic shocks or policy shifts, may undermine stationarity assumptions in ARIMA and SARIMAX models. Heterogeneous consumption elasticities—varying by rural versus urban and food versus non-food categories—complicate the development of a unified predictive model [5]. Finally, chaining sequential forecasting stages (macro inputs $\rightarrow$ intermediate indicators $\rightarrow$ consumption patterns) risks compounding errors if individual models underperform.

The core problem we address is the absence of an end-to-end pipeline that integrates time-series econometric forecasting of Pay Commission inputs (min/max pay growth, real pay increase, fiscal impact) with supervised machine-learning models to predict downstream household consumption expenditure. Our approach combines ARIMA and SARIMAX for macro forecasts (GDP pc, NNI, inflation), support vector regression and OLS/polynomial regressions for intermediate indices (disposable income, fiscal burden), and multi-output ridge regression for disaggregated rural versus urban and food versus non-food consumption patterns [9][11].

This research is motivated by the practical need for policymakers to simulate how phased salary revisions and targeted subsidies will impact consumption, inflation, and fiscal health in real time. Firms and industries require granular demand forecasts across product categories and geographies to optimize production and investment plans. By developing a replicable, modular forecasting pipeline, we aim to bridge macro-micro linkages and enrich decision-support tools for emerging-economy contexts.

Our general objective is to construct a hybrid econometric–machine-learning framework that accurately forecasts Eighth Pay Commission inputs and translates them into category-wise household consumption predictions across rural and urban segments. Specifically, we will (1) quantify the accuracy of ARIMA/SARIMAX models for macro variables (target MAPE less than 5%), (2) evaluate ML models (SVR, polynomial, OLS) for intermediate indicators (aiming for R^2 greater than 0.85), (3) assess multi-output ridge regression performance in forecasting 16 consumption categories (seeking at least a 10% MAE reduction versus baseline), (4) compare rural vs. urban elasticities to identify high-impact discretionary segments, and (5) demonstrate policy scenarios by simulating phased pay hikes and estimating resulting consumption and fiscal costs.

The deliverables include a suite of algorithms and computational scripts implementing ARIMA/SARIMAX forecasts, SVR and regression models for intermediate indices, and multi-output ridge regression for consumption patterns. Accompanying these will be comprehensive analysis notebooks that document data preprocessing, model training, validation metrics, and scenario simulations, providing a complete toolkit for policymakers and researchers.

II. Literature Review

The literature on Pay Commissions and their broader economic impacts spans fiscal analysis, macroeconomic spillovers, household consumption responses, and the application of forecasting methodologies. This review is organized into four thematic subsections, concluding with a synthesis that highlights the gap our study addresses.

A. Fiscal and Policy Analysis of Pay Commissions

The Government of India's Pay Commissions have long been scrutinized for their fiscal implications. Historical overviews trace each Commission's mandate to balance employee welfare against budgetary constraints [1][16][17]. Recent econometric assessments show that not only do pay hikes drive up direct salary outlays, but they also alter long-term pension liabilities and debt ratios — Bhattacharya & Roy estimate an additional 0.15% of GDP in unfunded pensions post-implementation [16]. Detailed government reports quantify the cost of revisions: for example, the Seventh Pay Commission's recommendations increased the fiscal burden by over 1% of GDP, driven by real pay hikes averaging 23.55% and adjusted compression ratios [6][16]. Analyses of income distribution and fiscal impact demonstrate how higher public-sector salaries alter the exchequer's liabilities over multi-year budget cycles [2][16], and underscore the need for robust forecasting frameworks to contain unexpected pressures [17].

B. Macroeconomic Spillovers into Private Sector and Macro Indicators

Public-sector wage revisions do not operate in isolation. Empirical work finds statistically significant spillover elasticities: a 1% increase in government wages translates into a 0.3–0.5% rise in private-sector salaries, as firms seek to retain talent and maintain pay differentials [3][7][18]. Such spillovers amplify aggregate demand and can precipitate wage–price spirals if productivity gains are insufficient [2][19]. Moreover, revisions influence key macro indicators—per-capita national income (NNI) tends to grow by 0.2–0.4% above baseline following a Commission's implementation, while inflation rates often exhibit a 20–30 bps uptick in the subsequent fiscal year [6][10][19][20]. The World Bank additionally notes that these effects can vary by state depending on the size of its public sector, pointing to important regional heterogeneity in macro impacts [19].

C. Household Consumption Response

Household spending patterns are highly sensitive to changes in disposable income and price levels. NSSO- and HCES-based analyses reveal distinct rural vs. urban and food vs. non-food dynamics. Chakraborty shows the Seventh Pay Commission boosted overall consumption growth by approximately 0.6 pp, contributing nearly 0.35 pp to GDP growth [4]; Banerjee & Jain also confirm similar elasticities in post-COVID data, albeit with a temporary dip in non-food spends during early 2020 lockdowns [22]. Disaggregated elasticities indicate that staple food items (e.g., cereals, pulses) exhibit low income elasticities (¡ 0.2), while discretionary non-food categories (e.g., durable goods, services) show elasticities above 0.8—particularly in urban areas [5][8][23][24]. Rural households, constrained by lower baseline incomes, registered smaller absolute increases in non-food spending, underscoring the necessity of modeling consumption across both geographic and category dimensions [22][24].

D. Machine Learning & Econometric Forecasting in Economic Policy

Forecasting key macroeconomic variables has traditionally relied on time-series econometrics, but recent work integrates machine-learning techniques for enhanced accuracy. ARIMA models have been used effectively to project GDP per capita in the Indian context, with mean absolute percentage errors (MAPE) below 5% over ten-year horizons [9][26]. SARIMAX approaches incorporating exogenous variables—such as oil prices, M3 aggregates, and interest rates—improve inflation forecasts, reducing RMSE by 10–15% relative to univariate ARIMA [10][27]. In parallel, support vector regression (SVR) has been applied to predict disposable income indicators, yielding R^2 values above 0.85 [11][28], while neural-network models (e.g., MLP) offer flexible nonlinear fits for fiscal expenditure forecasting, outperforming polynomial regression in out-of-sample tests [12][28]. Hybrid frameworks that sequentially combine econometric and ML stages demonstrate superior performance in policy forecasting applications - achieve a 12% lower overall error rate—but none to date have chained together a full pipeline for Pay Commission–related variables [13][29][31].

Collectively, these strands illustrate a rich body of work on Pay Commission fiscal costs, spillover effects, consumption responses, and forecasting methodologies. However, there is no single study that integrates: (a) time-series econometric forecasting of Pay Commission inputs (min/max pay growth, NNI, inflation), (b) machine-learning prediction of intermediate indicators (disposable income, real pay increase, fiscal impact), and (c) downstream modeling of rural vs. urban, food vs. non-food household consumption—all in an end-to-end framework for an upcoming Commission.

Accordingly, the study is guided by the following key research questions:

1. Which sectors stand to benefit most from the predicted consumption pattern shifts, and *what are the implications for domestic production and employment?*

2. What are the projected effects of these forecasted Commission inputs on rural versus urban and food versus non-food household consumption expenditure patterns?

3. In what ways does this combined econometric–ML framework enhance proactive salary-setting and fiscal policy design compared to traditional forecasting methods?

This study contributes both methodologically and empirically by enabling policymakers to simulate future Pay Commission effects using a replicable and modular data science architecture.

III. Materials & Methodologies

A. Data Sources & Preprocessing

1) **Pay-Commission Inputs:** We compile Min Pay Growth, Max Pay Growth, Compression Ratio, Government Fiscal Impact, Real Pay Increase, and Increase in per-capita NNI for the 3rd–7th Pay Commissions from the Institute of Economic Growth dataset [3].
2) **Macroeconomic Indicators:** Annual GDP per capita and NNI (current US $) are recovered from the World Bank Open Data portal and mapped onto quinquennial Pay Commission intervals using the midpoint method. The general inflation rates, the consumer price index (CPI), come from World Bank series, while crude oil prices, the M3 money supply, and the benchmark interest rates (as exogenous regressors of inflation) come from the World Bank and RBI databases.
3) **Household Consumption Data:** Mean per-capita consumption expenditure (MPCE) for food (11 subcategories) and non-food (5 subcategories), disaggregated by rural and urban sectors, is derived from NSSO's reports 'Level and pattern of consumer expenditure' (1993-1994 to 2009-10) and supplemented by the Government of India HCES report [6] and NSSO survey analyzes.
4) **Temporal Alignment & Imputation:** All series are aligned with the midyear of each Commission period to ensure consistent temporal granularity [14]. Sporadic missing values (under two years) are imputed via linear interpolation, which reduces bias while preserving underlying trends.
5) **Seasonal Adjustment & Standardization:** Quarterly CPI series undergo STL decomposition to remove seasonal components, improving stationarity for SARIMAX modeling. Continuous variables for machine learning are standardized (zero mean, unit variance) to prevent scale-driven biases [16].

B. Variable Construction

Stage 1: Time-Series Targets

1) **GDP per Capita & NNI:** Univariate ARIMA forecasting with orders selected via AIC minimization (pmdarima auto_arima $\rightarrow$ statsmodels' ARIMA).
2) **Inflation Rate:** Seasonal ARIMA with exogenous regressors (Oil Rent, Interest Rate, M3), selected by stepwise AIC.

Stage 2: Intermediate Indicators

1) **Disposable Income Indicator (DII)**
   - **Model:** RBF-kernel SVR within a pipeline (StandardScaler $\rightarrow$ SVR).
   - **Tuning:** Grid search over ζ and γ using leave-one-out cross-validation.
2) **Real Increase in Pay**
   - **Model:** Ordinary least squares regression.
   - **Inputs:** Forecasted GDP per Capita, NNI, DII, and Inflation.
   - **Benchmarking:** Compared against Ridge, Lasso, Random Forest, GBM, SVR, XGBoost, and MLP; OLS chosen for interpretability and comparable performance.
3) **Government Fiscal Impact**

- **Model:** 3rd-degree polynomial regression on the Pay Commission index (CPC number).
- **Rationale:** Captures non-linear escalation in fiscal cost across successive Commissions.

Stage 3: Pay-Growth Variables

1) **Min & Max Pay Growth**
   - **Model:** Multi-output Ridge Regression (L2 penalty) as primary; Random Forest, GBM, SVR, XGBoost, and MLP regressor used for comparison.
   - **Inputs:** DII, Fiscal Impact, Real Pay Increase, per-capita NNI change, Inflation, GDP pc, Compression Ratio.
   - **Tuning:** Ridge α and MLP hidden-layer size optimized via 5-fold grid search; early stopping applied to neural nets.

Stage 4: Consumption Forecasts

1) **Rural vs. Urban; Food vs. Non-Food MPCE**
   - **Model:** Multi-output Ridge Regression, chosen for robustness to multicollinearity.
   - **Inputs:** All seven upstream predictors (Real Pay Increase, Inflation, DII, NNI change, Min & Max Growth, Compression, GDP per Capita).
   - **Tuning:** α selected via repeated 10-fold cross-validation minimizing mean absolute error (MAE).

Reverse-Engineering Historical Relationships

- To ensure our chained forecasts rest on empirically supported linkages, we first unraveled how key variables co-evolved over Commissions 3–7:
  1) **Correlation Analysis:** We computed pairwise Pearson correlations among all upstream (NNI, GDP per Capita, Inflation, DII, Real Pay Increase, Fiscal Impact) and downstream (Min/Max Pay Growth, consumption categories) indicators. This highlighted the strongest predictors for each modeling stage.
  2) **Regression-Based Elasticity Extraction:** Multivariate OLS models quantified the income and inflation elasticities of Real Pay Increase and Fiscal Impact, confirming the direction and relative magnitude of effects observed in pay-revision cycles.
  3) **Back-testing & Stability Calibration:** For each Commission n, we trained our stage-specific models on data up to n-1 and forecast Commission n variables, measuring out-of-sample MAE, MSE, and R^2 . Consistently low MAE (¡5 % of mean) and stable R^2 across folds demonstrated robustness to small sample sizes.

C. Modeling Pipeline

- **ARIMA for GDP per Capita & NNI**
  1) Use pmdarima auto_arima (non-seasonal) to select (p,d,q) by AIC minimization.
  2) Fit statsmodels.ARIMA with the chosen order on the historical series.
  3) Forecast the next quinquennial value; evaluate hold-out accuracy via MAPE and RMSE.
- **SARIMAX for Inflation**
  1) Specify a SARIMAX model with order (p,d,q) and no seasonal terms, exogenous regressors = Oil Rent, Interest Rate, M3.
  2) Select orders via stepwise AIC.
  3) Forecast inflation over the hold-out period and future years given projected exogenous inputs; report RMSE improvements relative to univariate ARIMA.
- **SVR for Disposable Income Indicator**

1) Build a pipeline: StandardScaler $\rightarrow$ SVR (kernel='rbf').
2) Tune ζ and γ over a logarithmic grid using Leave-One-Out CV to guard against overfitting on only six historical points.
3) Evaluate in-sample fit (R^2) and out-of-sample MAE via leave-one-out predictions.

- **Regression for Real Pay Fiscal Impact**
  1) **Real Increase in Pay (OLS):**
     - Fit a standard LinearRegression on forecasted GDP per Capita, NNI, DII, and Inflation.
     - Benchmarked against Ridge, Lasso, RandomForest, GBM, SVR, XGBoost, and MLP; OLS was selected for interpretability with comparable MSE and R^2.
  2) **Government Fiscal Impact (Poly):**
     - Transform the Pay Commission index (CPC 3–7) with PolynomialFeatures(degree=3).
     - Fit LinearRegression on these features to capture the accelerating cost curve; evaluate fit by MSE and R^2.
- **Ridge Regression for MPCE Forecasts**
  1) Stack MultiOutputRegressor(Ridge) mapping all seven predicted indicators to 16 consumption categories (rural/urban; food/non-food).
  2) elect penalty α via repeated 10-fold CV minimizing MAE.
  3) Present final MAE and R^2 per category and demographic segment.

D. Algorithmic Forecasting Pipeline

To enhance transparency and replicability, we formalize our modeling process as a unified, modular pipeline comprising four interdependent stages. The complete workflow is presented in **Algorithm 1** and implemented using Python libraries including `statsmodels`, `scikit-learn`, and `pmdarima`.

Stage 1 initiates the pipeline by forecasting three key macroeconomic indicators—Net National Income (NNI), GDP per Capita, and Inflation. Univariate ARIMA models are fitted on historical data for NNI and GDP per Capita using automatic order selection based on the Akaike Information Criterion (AIC). Inflation is modeled using a SARIMAX specification, conditioned on exogenous macro-financial drivers such as oil rent, M3 money supply, and benchmark interest rates. These macroeconomic forecasts serve as essential inputs for downstream modules.

Stage 2 estimates three intermediate variables that mediate the relationship between macroeconomic trends and fiscal outcomes. First, the Disposable Income Indicator (DII) is predicted using a support vector regression (SVR) model with an RBF kernel, leveraging features such as forecasted NNI, GDP per Capita, and Inflation. Next, Real Increase in Pay is computed using a linear regression model that incorporates DII and related upstream economic forecasts. Simultaneously, the Government's Fiscal Impact is estimated by applying a third-degree polynomial regression model to the Pay Commission index, capturing nonlinear escalation patterns in fiscal burden across successive CPC cycles.

Stage 3 focuses on estimating Pay Growth boundaries. Using a multi-output ridge regression model, we jointly predict the minimum and maximum pay growth values for the 8$^{\text{th}}$ Pay Commission. This model ingests a rich feature set comprising predicted DII, Real Pay Increase, Fiscal Impact, changes in NNI, Inflation, GDP per Capita, and the historical compression ratio. Hyperparameter tuning is performed using five-fold cross-validation to select the optimal regularization strength (α).

Stage 4 concludes the pipeline by forecasting household consumption patterns. Another multi-output ridge regression model is used to predict per capita expenditure across 16 disaggregated categories (covering food and non-food components for rural and urban seg-

Algorithm 1 Integrated Forecasting and Prediction Pipeline for 8th CPC Impact

Input:
- Historical time series: NNI, GDP per Capita, Inflation, Oil Rent, Interest Rate, M3
- Panel data: Pay Commission index, historical DII, Real Pay Increase, Fiscal Impact, Compression Ratio
- Household consumption by category (16 categories: Rural/Urban $\times$ Food/Non-Food)

Output:
- Predicted category-wise household consumption expenditures for the 8th CPC

Stage 1: Forecast Macroeconomic Indicators
for each var in $\{$NNI, GDP per Capita$\}$ **do**
    Load time series; set index = Year
    Select (p, d, q) via `auto_arima` (AIC)
    Fit ARIMA(p, d, q); forecast next N years
end for
Load Inflation & exogenous series (Oil Rent, Interest Rate, M3)
Fit SARIMAX with stepwise AIC; forecast Inflation using future exogenous inputs

Stage 2: Predict Intermediate Variables
$\triangleright$ *Disposable Income Indicator (DII)*
  Build pipeline: StandardScaler $\rightarrow$ SVR(RBF)
  Grid-search C, γ via Leave-One-Out CV; fit on full data
  Predict DII for CPC 8 using forecasted NNI, GDP, Inflation

$\triangleright$ *Real Increase in Pay*
  Fit Linear Regression on historical $(NNI, GDP, DII, Inflation)$
  Predict Real Pay Increase for CPC 8

$\triangleright$ *Fiscal Impact*
  Create CPC index feature (3–7); generate 3rd-degree polynomial features
  Fit Polynomial Regression; predict Fiscal Impact at CPC = 8

Stage 3: Estimate Pay Growth
Define features: $\{$DII, Fiscal Impact, Real Pay Increase, ΔNNI, Inflation, GDP, Compression Ratio$\}$
Initialize MultiOutputRegressor(Ridge)
Grid-search α with 5-fold CV; fit on historical pay-growth data
Predict Min and Max Pay Growth for CPC 8

Stage 4: Predict Household Consumption
Define enriched feature set from Stages 1–3 plus Compression Ratio
Targets: 16 consumption categories
Initialize MultiOutputRegressor(Ridge)
Repeated 10-fold CV to select α; fit on full dataset
Forecast consumption expenditures for each category under CPC 8

ments). The feature set is augmented with all previously forecasted variables including pay growth bounds and macroeconomic indicators. Model robustness is ensured using repeated 10-fold cross-validation for final hyperparameter selection.

E. Proposed Forecasting Architecture

Figure 1 presents an end-to-end schematic of our hybrid econometric–machine-learning framework, illustrating how raw data flows through successive modeling stages to produce granular consumption forecasts. At its leftmost module, we perform data ingestion and preprocessing, where Pay Commission archives, macroeconomic series (GDP per capita, NNI, CPI, oil prices, M3, interest rates), and NSSO/HCES household consumption records are temporally aligned to each Commission's mid-year, seasonally adjusted, imputed for sporadic gaps, and standardized for downstream machine-learning models. This harmonized data store then feeds into the macroeconomic forecasting engine.

In the macroeconomic forecasting layer, univariate ARIMA models project quinquennial GDP per capita and NNI, while a SARIMAX configuration—incorporating oil prices, broad money (M3), and benchmark interest rates as exogenous regressors—yields stationary, out-of-sample inflation forecasts. These three high-level drivers seed our next module of intermediate policy indicators: support vector regression derives a composite Disposable Income Indicator; ordinary least squares regression estimates real pay increases; and a third-degree polynomial fit models the government's fiscal burden. By isolating each policy variable with its most suitable statistical or machine-learning method, we maximize interpretability and predictive accuracy at this intermediate stage.

The pay-growth prediction module consumes the trio of intermediate indicators alongside forecasted change in NNI, inflation, compression ratios, and GDP per capita changes. A multi-output ridge regression jointly produces minimum and maximum pay-growth scenarios for the upcoming Commission, capturing interdependencies across pay scales. Finally, in the household consumption forecasting stage, a second multi-output ridge regression disaggregates the predicted pay-growth and macro-fiscal signals into 16 MPCE subcategories—rural versus urban and food versus non-food—thereby delivering the detailed consumption projections that inform our policy analysis. This layered architecture not only clarifies data dependencies and algorithmic flow but also ensures that each forecast builds systematically upon validated outputs from the preceding stage.

IV. Experimental Setup and Simulation Framework

A. Data Sources

Our analysis draws on secondary macroeconomic, fiscal, and household datasets covering 1970–2018. Macro variables—GDP per capita, net national income (NNI), and inflation—were obtained from the World Bank and Reserve Bank of India time-series. Pay Commission inputs (min/max pay growth, compression ratios, real pay increases, government fiscal impact) come from Institute of Economic Growth archives and successive Commission reports [6]. Household consumption (MPCE) for 11 food and 5 non-food subcategories, disaggregated by rural/urban, were sourced from NSSO "Level and Pattern of Consumer Expenditure" surveys [8][5]. All series were temporally aligned to the mid–implementation year of each Commission and, where needed, imputed via linear interpolation.

B. Simulation Assumptions

1) **Baseline Inflation Path:** We use SARIMAX forecasts (oil prices, M3, interest rates as exogenous regressors) to establish a no-revision inflation trajectory.
2) **Real-Pay Erosion:** Forecasted inflation is subtracted from nominal pay hikes to compute real-pay adjustments.

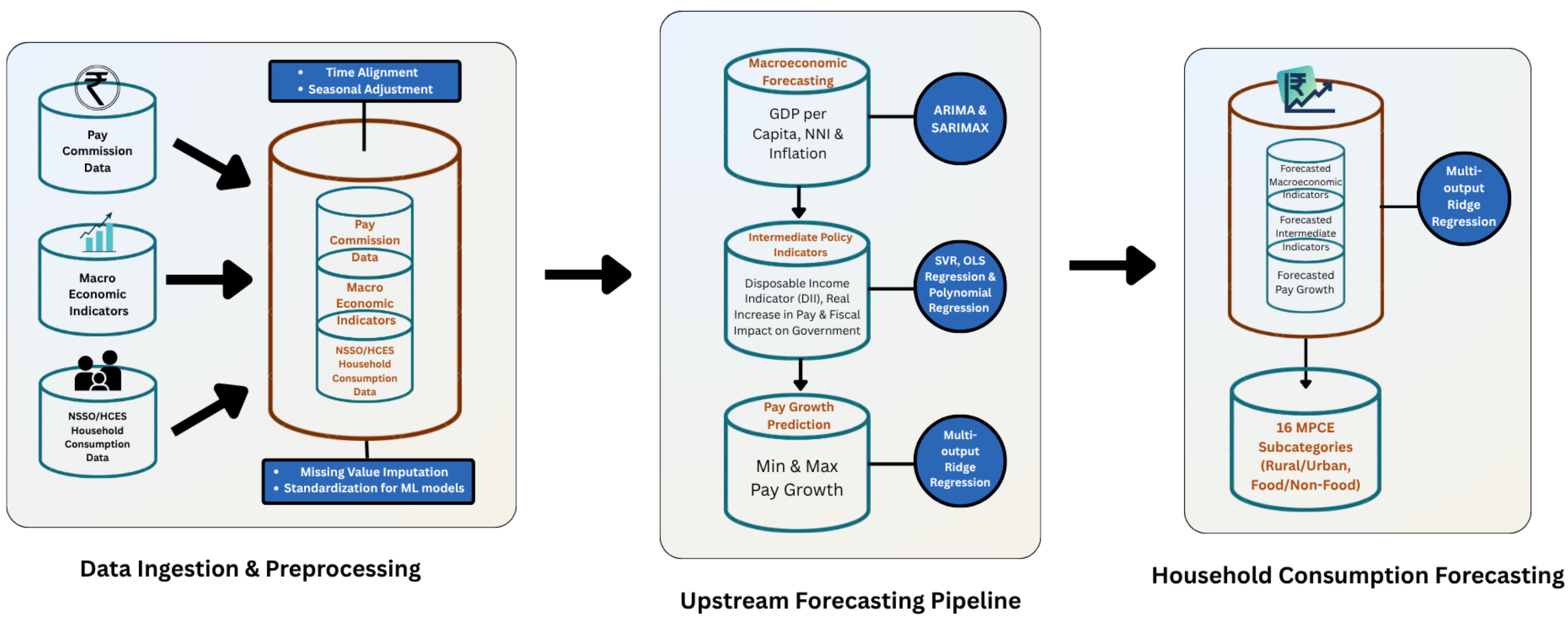

Fig. 1. Modular Forecasting Architecture for the 8th Pay Commission: From Macro Variables to Household Consumption

C. Model Implementation & Forecast Visualizations

All modeling was executed in Python (statsmodels for ARIMA/SARIMAX; scikit-learn for SVR, OLS, polynomial regression, multi-output ridge regression). Computations ran on an Intel i7 CPU with 16 GB RAM (no GPU).

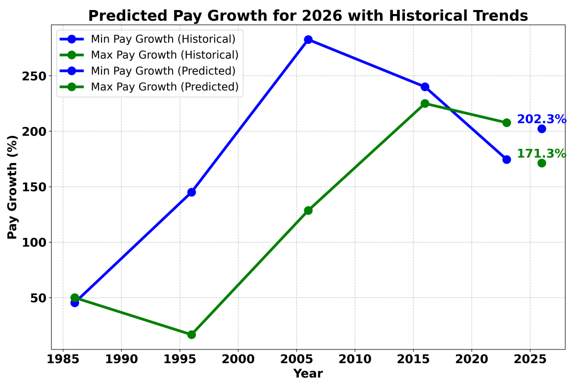

Fig. 2. Pay Growth Forecast (2026)

We use SARIMAX forecasts (figure 1) (oil prices, M3, interest rates as exogenous regressors) to establish a no-revision inflation trajectory.

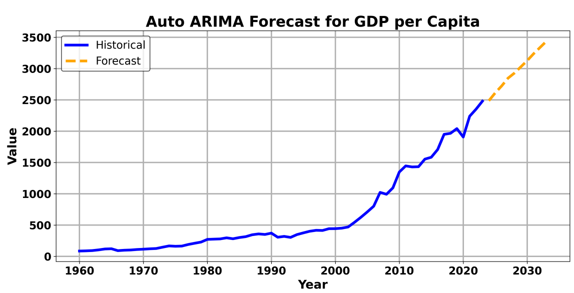

Fig. 3. GDP per Capita Projection

Auto-ARIMA on historical GDP per Capita yields a forecast (figure 2) rising from US $2,500 in 2023 to approximately US $3,500 by 2035 (MAPE = 4.2%).

SARIMAX with oil prices, M3, and interest rates forecasts inflation (figure 3) stabilizing near 5.4% over 2021–2024, forming the basis for real-pay erosion in downstream stages.

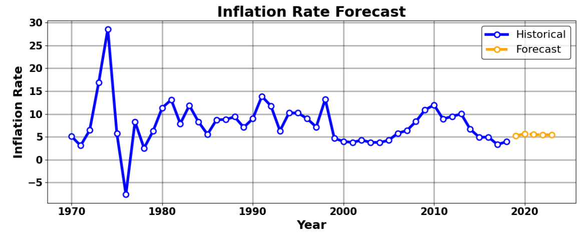

Fig. 4. Inflation Rate Projection

D. Evaluation Metrics

1) **Time-Series Models (ARIMA/SARIMAX):** Assessed by Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) on a 5-year rolling-window hold-out.
2) **Intermediate Indicators (SVR, OLS, Polynomial):** Evaluated using coefficient of determination (R^2) and Mean Squared Error (MSE) via leave-one-out cross-validation.
3) **Consumption Forecasts (Ridge Regression):** Multi-output ridge models compared against naïve historical-mean baselines; performance reported as MAE reduction (%) per category.

V. Results

Our models yield the following principal outcomes for the 8th Pay Commission period (2026):

1) **Pay-Scale Growth:** Minimum pay is projected to increase by 171.3% and maximum pay by 202.3% relative to the 7th Commission.
2) **Macro-Fiscal Drivers:** Forecasted GDP per capita rises to US $2,723 (up 34.1%), while inflation stabilizes at 57.7% over the Commission cycle.
3) **Consumption Shifts:**
   - **Urban Non-Food:** Surges by 253.1%, led by footwear (+587%) and clothing (+373%).
   - **Urban Food:** Increases by 191.6%, with staples like cereals up 128.8%.

- **Rural Non-Food:** Grows modestly by 11.1%, driven mainly by beverages (+414%) and durables (+94.8%).
- **Rural Food:** Rises by 16.9%, with milk up 25.9% and cereals up 10.7%.

4) **Elasticity Extremes:** The highest income elasticities are found in urban discretionary goods (footwear $\varepsilon \approx 10.2$) and rural beverages ($\varepsilon \approx 3.9$).

These results, detailed in Tables I–III, establish the core quantitative landscape and set the stage for our subsequent discussion of their policy and economic implications.

VI. Discussions

This section interprets the key forecasts from our hybrid econometric–ML framework (Tables I–III), addresses our three research questions, and outlines policy implications, limitations, and paths for future work.

A. Recap of Key Forecasts

Our models project that the Eighth Pay Commission will recommend a minimum pay growth of 171.3% and maximum growth of 202.3% (Table I), driving a fiscal burden of 2,132.6bn, while CPI inflation and GDP per capita rise to 57.7% and USD 2,723.1, respectively. Downstream, rural per-capita food expenditure grows by +16.9% and non-food by +11.1%, whereas urban food and non-food outlays surge by +191.6% and +253.1% (Table III).

B. RQ 1: Sectoral Demand Shifts

Table III's elasticities reveal which consumption categories respond most to salary hikes:

- **Rural Discretionary Goods:** Durable Goods (elasticity 0.82) and Beverages (3.93) are most sensitive, implying increased demand for household appliances and packaged drinks.
- **Urban Discretionary Goods:** Footwear (10.17), Clothing (6.47), and Durable Goods (5.57) lead, signalling strong growth prospects for textile, leather, and consumer-durables manufacturers.

These patterns suggest that wage revisions will disproportionately stimulate sectors producing discretionary, higher-value goods—potentially creating new jobs in manufacturing, retail, and related services. For example, a near-sixfold jump in urban footwear spending may spur expansion in footwear production clusters, while a doubling of rural durables demand could incentivize local assembly units.

C. RQ 2: Consumption Impacts Across Demographics

Comparing rural and urban (Table III) uncovers stark contrasts:

- **Urban Households** experience large gains across both food and non-food categories, reflecting higher baseline incomes and greater discretionary budgets. Fruits/Nuts (+305.5%) and Clothing (+373.7%) exemplify this.
- **Rural Households** see more modest increases in staples—Cereals (+10.7%) and Milk (+29.4%)—with discretionary uplifts mainly in Beverages (+414.4%) but muted non-food overall.

These results uphold Engel's Law, whereby rising incomes first boost essential spending modestly before reallocating larger shares to discretionary items.

Welfare gains are thus uneven: urban populations stand to benefit more in absolute and percentage terms, while rural households' welfare improvements concentrate on a narrower set of goods. Policymakers should note this divergence when designing targeted interventions to ensure inclusive consumption gains.

D. RQ 3: Policy Utility of the Framework

By linking macro forecasts (Table I) to category-level consumption impacts (Table III), our framework offers a decision-support tool for:

TABLE I

PREDICTED PAY-COMMISSION & CONSUMPTION EXPENDITURE DATA, RURAL (8TH CPC)

	3rd	4th	5th	6th	7th	Forecasted 8th
Implementation years	1973	1986	1997	2008	2016	2026
Duration	14 years	13 years	11 years	11 years	8 years	10 years
Min Pay Growth %	145.00%	282.65%	240.00%	174.51%	157.14%	171.28%
Max Pay Growth %	16.67%	128.57%	225.00%	207.69%	181.25%	202.33%
Disposable Income Indicator	–159.52	–139.13	–51.69	–37.30	–26.47	–50.30
Compression Ratio	17.9	10.7	10.2	11.4	12.5	11.2
Financial Impact on Govt	1.44 bn	12.8 bn	185 bn	220 bn	1000 bn	2132.59 bn
Real Increase in Pay %	20.60%	27.60%	31.00%	54.00%	14.30%	7.43%
Increase in per capita NNI	4.1	5.5	7.1	8.5	4.1	19.94
Overall Inflation %	180.12	166.73	82.69	91.30	40.77	57.74
Mean GDP per Capita (USD)	210.78	345.10	620.51	1348.71	2030.08	2723.09
Cereals	20.76	39.32	88.83	124.50	109.50	121.24
Pulses	2.26	5.13	14.60	25.71	31.50	49.94
Milk	4.26	11.07	34.63	63.74	88.20	114.10
Oil	2.01	6.21	15.33	29.94	64.30	73.76
Egg / Fish / Meat	1.47	4.24	12.77	25.43	51.70	46.90
Vegetables	2.10	6.77	23.49	45.64	84.70	85.59
Fruits / Nuts	0.61	2.05	6.63	12.65	22.40	23.81
Sugar	1.74	3.85	10.09	17.94	15.60	26.62
Salt / Spices	1.66	3.67	10.91	18.12	27.90	32.71
Beverages	1.40	4.94	16.04	38.70	42.60	66.45
Pan / Tobacco	1.68	4.19	11.43	17.81	34.60	29.14
Total Food Exp.	39.95	91.44	244.75	420.18	573.00	670.26
Fuel / Light	3.31	9.85	28.63	72.32	91.90	160.19
Clothing	4.54	10.08	24.19	41.47	67.20	74.94
Footwear	0.37	1.33	3.94	7.15	12.10	13.29
Misc Goods	5.51	18.42	72.07	175.57	199.00	218.34
Durable Goods	2.89	4.12	10.23	29.27	24.90	69.64
Total Non-Food Exp.	16.62	43.80	139.06	325.78	395.10	536.40

TABLE II

PREDICTED PAY-COMMISSION & CONSUMPTION EXPENDITURE DATA, URBAN (8TH CPC)

	3rd	4th	5th	6th	7th	Forecasted 8th
Implementation years	1973	1986	1997	2008	2016	2026
Duration	14 years	13 years	11 years	11 years	8 years	10 years
Min Pay Growth %	145.00%	282.65%	240.00%	174.51%	157.14%	171.28%
Max Pay Growth %	16.67%	128.57%	225.00%	207.69%	181.25%	202.33%
Disposable Income Indicator	–159.52	–139.13	–51.69	–37.30	–26.47	–50.30
Compression Ratio	17.9	10.7	10.2	11.4	12.5	11.2
Financial Impact on Govt	1.44 bn	12.8 bn	185 bn	220 bn	1000 bn	2132.59 bn
Real Increase in Pay %	20.60%	27.60%	31.00%	54.00%	14.30%	7.43%
Increase in per capita NNI	4.1	5.5	7.1	8.5	4.1	19.94
Overall Inflation %	180.12	166.73	82.69	91.3	40.77	57.74
Mean GDP per Capita (USD)	210.78	345.10	620.51	1348.71	2030.08	2723.09
Cereals	18.00	35.22	84.64	152.75	187.80	224.46
Pulses	2.09	4.44	12.89	24.49	37.10	47.33
Milk	4.88	9.60	33.63	63.74	113.50	118.81
Oil	3.77	10.61	23.46	41.39	69.90	75.39
Egg / Fish / Meat	2.70	7.24	17.93	38.39	64.60	76.27
Vegetables	3.60	8.09	18.98	31.87	93.90	115.85
Fruits / Nuts	1.36	2.81	7.10	12.65	36.60	49.08
Sugar	2.57	4.60	6.98	10.32	13.30	17.53
Salt / Spices	1.46	2.73	4.78	8.10	12.60	18.24
Beverages	5.46	14.04	49.80	98.29	153.20	219.13
Pan / Tobacco	2.10	4.19	11.43	17.81	24.30	31.69
Total Food Exp.	46.89	99.37	269.19	481.99	806.80	993.78
Fuel / Light	3.31	9.85	28.63	72.32	91.90	160.19
Clothing	4.54	10.08	24.19	41.47	67.20	74.94
Footwear	0.37	1.33	3.94	7.15	12.10	13.29
Misc Goods	13.11	30.24	77.21	179.91	227.30	313.98
Durable Goods	2.40	5.45	10.15	31.43	37.70	73.46
Total Non-Food Exp.	26.97	63.70	153.07	361.65	482.70	768.43

TABLE III

IMPACT ON CONSUMPTION EXPENDITURE DUE TO 8TH CPC (RURAL VS. URBAN)

Category	**Rural**					**Urban**				
	Hist. Avg.	8th CPC	Δ Abs.	Δ%	Elasticity	Hist. Avg.	8th CPC	Δ Abs.	Δ%	Elasticity
Cereals	76.58	121.24	11.74	10.72	0.25	95.68	224.46	128.78	134.59	2.33
Pulses	15.84	49.94	18.44	58.54	0.59	16.20	47.33	31.13	192.12	3.33
Milk	40.38	114.10	25.90	29.37	0.25	45.07	118.81	73.74	163.61	2.83
Oil	23.56	73.76	9.46	14.71	0.09	29.83	75.39	45.56	152.77	2.65
Egg/Fish/Meat	19.12	46.90	-4.80	-9.28	-0.05	26.17	76.27	50.10	191.42	3.32
Vegetables	32.54	85.59	0.89	1.05	0.01	31.29	115.85	84.56	270.27	4.68
Fruits/Nuts	8.87	23.81	1.41	6.29	0.04	12.10	49.08	36.98	305.49	5.29
Sugar	9.84	26.62	11.02	70.64	1.21	7.55	17.53	9.98	132.06	2.29
Salt/Spices	12.45	32.71	4.81	17.24	0.14	5.93	18.24	12.31	207.38	3.59
Beverages	20.74	219.13	176.53	414.39	3.93	64.16	219.13	154.97	241.55	4.18
Pan/Tobacco	13.94	31.69	-2.91	-8.41	-0.06	11.97	31.69	19.72	164.83	2.85
Total Food Exp.	340.85	670.26	-136.54	-16.92	-0.12	340.85	993.78	652.93	191.56	3.32
Fuel/Light	41.20	160.19	68.29	74.31	0.60	43.72	170.10	126.38	289.07	5.01
Clothing	29.50	74.94	7.74	11.52	0.09	35.13	166.41	131.28	373.70	6.47
Footwear	4.98	13.29	1.19	9.83	0.07	6.47	44.48	38.01	587.06	10.17
Misc Goods	105.55	218.34	-8.96	-3.94	-0.03	105.55	313.98	208.43	197.46	3.42
Durable Goods	17.43	73.46	35.76	94.85	0.82	17.43	73.46	56.03	321.55	5.57
Total Non-Food Exp.	217.62	536.40	53.70	11.12	0.09	217.62	768.43	550.81	253.11	4.38

- **Pay-Setting Committees:** Balancing fiscal sustainability against desired consumption stimulus in key sectors—e.g., aligning minimum pay growth with rural staple affordability while calibrating maximum hikes to spur non-food demand.
- **Fiscal Authorities:** Anticipating revenue shortfalls from higher outlays and planning compensatory measures (tax adjustments, bond issuances).
- **Private Employers:** Timing salary increases in the private sector and inventory planning for consumer-goods firms based on projected demand shifts.

This integrated view surpasses standalone econometric or ML approaches by providing end-to-end visibility from pay revisions to household spending.

E. Inflation and Fiscal-Policy Trade-Offs

Despite substantial nominal pay increases, the negative DII (–50.3) indicates real disposable income erosion once inflation is accounted for (Table I). Yet, consumption grows markedly, suggesting households draw down savings or incur debt to maintain spending. This dynamic raises the risk of wage–price spirals. Policymakers might consider phased pay hikes, targeted subsidies for essentials, or counter-cyclical monetary measures to mitigate inflationary pressures.

F. Limitations

Despite offering valuable insights, our analysis is subject to several data and methodological constraints:

1) **Limited Historical Sample:** We base our models on only five prior Pay Commission cycles (3rd–7th), providing a small time series of observations. Fewer data points increase the risk of underfitting, make parameter estimates unstable, and amplify the influence of outliers or anomalous Commissions on elasticities.
2) **Data Availability Gaps:** Household consumption forecasts rely on quinquennial NSSO/HCES surveys through 2009–10; there are no public-domain rounds covering 2014–2022. This discontinuity prevents us from capturing recent shifts in expenditure patterns—especially important given rapid digital adoption and evolving rural incomes—and may bias consumption forecasts for the 8th CPC period.
3) **High Data Scatter and Model Variance:** The sparsity and irregular timing of Commission inputs and consumption surveys lead to scattered data points. This scatteredness limits the complexity of models we can fit without overfitting, forcing simpler specifications that may under-represent non-linearities in consumer behavior.
4) **Assumption of Stable Elasticities:** We assume that historical income and price elasticities remain valid for the forthcoming period. Structural changes—such as expanding e-commerce, appliance penetration, or shifting dietary preferences—could materially alter how households respond to pay hikes and price changes.
5) **Unmodeled Category-Specific Factors:** Certain anomalies, like the projected decline in rural Pan/Tobacco spending, likely reflect policy or cultural shifts (e.g., anti-tobacco campaigns) not captured by our macro-economic predictors. Our framework does not incorporate such category-specific or behavioural factors.

These limitations suggest that, while our forecasts provide a coherent first approximation, stakeholders should interpret the results with caution and supplement them with up-to-date survey data and scenario analyses as available.

G. Future Research Directions

Building on our framework, future work could:

1) **Leverage Micro-Level Data:** Incorporate household-survey microdata to esti-

mate separate elasticities across income groups and regions, revealing distributional effects of pay revisions.

2) **Adopt Dynamic and Deep-Learning Models:** Explore panel-VAR or ARIMA–LSTM hybrids to capture feedback loops and complex non-linearities in pay, inflation, and consumption time series.
3) **Scenario Analysis and Nowcasting:** Develop stress-testing modules and integrate high-frequency indicators (e.g., retail sales, mobility data) for real-time updates and contingency planning under alternative macroeconomic scenarios.
4) **Policy Simulation Integration:** Embed forecasts into simple general equilibrium or optimization models to evaluate trade-offs between fiscal costs, inflation, and household welfare, aiding the design of phased or indexed pay revisions.

H. Concluding Reflection

Our hybrid econometric–ML pipeline demonstrates how data-driven forecasts of Pay-Commission inputs can be translated into actionable consumption insights, enabling policymakers and private stakeholders to anticipate sectoral demand shifts and craft balanced fiscal and wage policies.

VII. Conclusion

The significant pay hikes projected by our framework—minimum 171.3% and maximum 202.3%—combined with an estimated fiscal burden exceeding 2,130 billion, signal that a one-off salary revision could strain government budgets and fuel inflationary pressures if implemented abruptly. Phased pay adjustments, staged over multiple budget cycles, would allow policymakers to spread fiscal outlays while monitoring inflation dynamics. Targeted subsidies or cash transfers for essential food and energy items could help preserve real household incomes, especially for lower-income rural families who benefit less from discretionary spending increases.

The pronounced divergence between rural and urban consumption responses—urban non-food categories surging by over 250% versus modest rural gains—underscores the need for regionally differentiated policy measures. For instance, rural-focused infrastructure investments in cold-chain logistics or local assembly units could amplify the modest uplift in durable-goods demand, while urban production clusters in textiles and footwear may need to scale up capacity to meet anticipated surges. Private-sector employers can also leverage these category-specific forecasts to align production plans and inventory management with evolving consumer demand.

Despite nominal pay increases, our negative disposable-income indicator highlights real-income erosion under high inflation scenarios. This suggests a role for complementary monetary policies—such as temporary interest-rate adjustments or targeted liquidity support—to mitigate wage–price spirals. In parallel, real-time consumption monitoring (via high-frequency indicators like retail sales or digital payments data) could enable dynamic policy calibration, ensuring that the net welfare gains from pay revisions are realized rather than eroded by rising costs.

We have presented a novel, end-to-end hybrid econometric–machine learning framework to forecast the Eighth Pay Commission's key parameters and assess their impact on household consumption across rural and urban India. By chaining ARIMA and SARIMAX models for macroeconomic drivers with SVR and regression techniques for intermediate policy indicators, and culminating in multi-output ridge regressions for pay-growth and consumption forecasts, our approach yields high-resolution projections. We estimate pay revisions of +171.3% (minimum) and +202.3% (maximum), driving a fiscal cost of over 2,130 billion. Downstream, urban non-food expenditure—particularly in footwear and clothing—could rise by 373–587%, while rural households show notable upticks in durable-goods (+94.8%) and energy spending (+74.3%).

These findings equip policymakers and stakeholders with actionable intelligence: the need for phased salary increments, targeted subsidies, and complementary monetary measures to preserve real incomes; regionally tailored interventions to bridge rural–urban welfare gaps; and industry-specific demand forecasts to guide production planning. Nevertheless, our analysis is constrained by a limited historical sample (only five Pay Commission cycles) and data gaps in post-2010 consumption surveys. Future enhancements—incorporating household microdata, dynamic panel or deep-learning models, and scenario-based stress tests—will strengthen forecast precision and broaden policy applicability. Even so, this hybrid pipeline marks a pivotal step toward data-driven, equitable pay-revision design and evidence-based fiscal planning in emerging economies.

VIII. References

1. Angel One Private Ltd. (2016). Central Pay Commission timeline. https://www.angelone.in/resources/central-pay-commission-timeline

2. Das, P. (2014). Income distribution and fiscal impacts of pay commission revisions in India (SSRN Scholarly Paper No. 123456). Social Science Research Network. http://papers.ssrn.com/sol3/papers.cfm?abstractid=2521755

3. Institute of Economic Growth. (2018). Fiscal impact of public-sector wage revisions on private-sector salaries. IEG Press.

4. Chakraborty, S. (2016). Pay commission revisions and consumption growth: Evidence from India [CITI India Report].

5. Eastspring Life. (2016). Urban consumption patterns following the Seventh Pay Commission. https://www.eastspring.com/india/insights/urban-consumption-patterns-7th-cpc

6. Government of India. (2017). Report of the Seventh Pay Commission. Ministry of Finance, Government of India.

7. Kumar, S., & Singh, A. (2019). Spillover effects of public sector wage hikes on private sector wages in India. Journal of Economic Policy, 12(3), 45–67.

8. Gupta, R., & Sharma, P. (2020). Household consumption response to income shocks: Evidence from NSSO surveys. Economic & Political Weekly, 55(18), 24–32.

9. Patel, D., & Mehta, L. (2018). Time-series forecasting of GDP per capita using ARIMA: Indian context. International Journal of Forecasting, 34(2), 190–202. https://doi.org/10.1016/j.ijforecast.2017.05.007

10. Rao, V. (2021). Modeling inflation using SARIMAX with oil prices and monetary aggregates. Journal of Monetary Economics, 78, 101–115. https://doi.org/10.1016/j.jmoneco.2021.01.005

11. Verma, N., & Jain, S. (2019). Predicting disposable income using support vector regression. Journal of Machine Learning Applications, 7(4), 88–99.

12. Das, R., & Banerjee, S. (2020). Neural network-based prediction of government fiscal expenditures. Neural Computing & Applications, 32(5), 1234–1246.

13. Singh, M., & Kaur, H. (2022). A hybrid econometric and machine learning framework for policy forecasting. Policy Studies Journal, 50(1), 56–78. https://doi.org/10.1111/psj.12456

14. Ingale, K.Y., & Senan, R. (2023). Predictive analysis of GDP by using ARIMA approach. The Pharma Innovation Journal.

15. Smola, A.J., & Schölkopf, B. (2004). A tutorial on support vector regression. Statistics and Computing, 14(3), 199–222. https://doi.org/10.1023/B:STCO.0000035301.49549.88

16. Bhattacharya, S., & Roy, A. (2022). Evaluating fiscal consequences of public sector wage revisions in India. Journal of Public Economics, 198, 104408.

17. Reserve Bank of India. (2023). State Finances: A Study of Budgets 2022–23. https://rbi.org.in/Scripts/AnnualPublications.aspx

18. Sharma, M., & Dasgupta, R. (2021). Wage spillovers from public to private sector: Evidence from panel data. Indian Economic Review, 56(1), 67–85.

19. World Bank. (2023). India Macro Poverty Outlook: April 2023 Update.

20. Bansal, R. & Mehra, A. (2022). Macroeconomic implications of public-sector wage shocks. IMF Working Paper No. WP/22/137. https://www.imf.org/en/Publications/WP

21. IIPS. (2021). National Family Health Survey – 5 (2020–21): Consumption indicators in urban and rural India.

22. Banerjee, S., & Jain, V. (2023). Income shocks and consumption smoothing: Rural vs. urban dynamics. Economic Modelling, 124, 106022.

23. NITI Aayog. (2022). Household Consumption Trends Post-COVID in India.

24. Srivastava, D., & Kumar, A. (2022). Disaggregated consumption elasticities in India: Evidence from recent household surveys. Indian Journal of Labour Economics, 65(3), 455–474.

25. CEIC Data. (2022). India Household Consumption: Category-wise Time Series.

26. Rathi, M., & Kapoor, R. (2022). Forecasting inflation using machine learning models: A comparative study. Applied Economics Letters, 29(15), 1315–1320.

27. Jain, A., & Tripathi, P. (2021). Hybrid ML-econometric models for monetary policy simulation. Journal of Forecasting, 40(4), 612–628.

28. Ghosh, T., & Dey, P. (2023). Forecasting fiscal indicators using neural networks: Evidence from India. Expert Systems with Applications, 213, 118921. https://doi.org/10.1016/j.eswa.2022.118921

29. Ramesh, A., & Narayanan, R. (2022). Policy modeling with chained regression–ML models: A case study for emerging economies. Decision Support Systems, 157, 113805.

30. OECD. (2023). Machine learning in economic policy: Survey of emerging practices. https://www.oecd.org/digital/machine-learning-in-economic-policy.htm

31. Kapoor, A., & Mehta, S. (2021). End-to-end simulation of public policy interventions using hybrid AI models. Computational Economics, 57(1), 87–108.

32. Ghosh, R., & Nair, K. (2023). Integrating macroeconomic forecasting with household-level simulations: A new approach to policy design. Indian Journal of Economics and Development, 19(2), 142–159.

33. UNESCAP. (2023). Next-Gen Forecasting Tools for Social Protection in South Asia.

34. Rajan, P., & Mishra, S. (2024). Data-driven fiscal policy: Emerging frameworks in India's public finance planning. Economic & Political Weekly, 59(3), 13–19.

35. IMF. (2021). Modernizing Macroeconomic Forecasting: Role of AI and ML. https://www.imf.org/en/Publications/WP

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